

Customer Intelligence

Executive Report

A 404-customer survey turned into a working business tool. Three complementary segmentation lenses · monthly-spend analysis · a what-if revenue projection · marketing playbook by segment.

Live dashboard: wine-shop-customer-dashboard-group3.streamlit.app

Executive Summary

The wine shop has an established but uneven customer base. Of the 404 customers we surveyed, an analytical minority drives the majority of revenue, the deli side of the business is materially under-monetised, and the data shows a consistent gender pattern in where customers drink — a pattern that translates directly into how the store should design campaigns and displays.

404	€2.78M	61%
customers surveyed	estimated annual revenue	of revenue from top 20% of customers

Three findings worth acting on now

- 1. Champions are the business.** 37% of customers — the Champions FM segment — generate 68% of estimated annual revenue. They visit weekly and spend above average. Anything that costs Champions to retain pays for itself many times over; this segment is the protection priority.
- 2. Loyal Regulars are the biggest upsell opportunity.** The largest segment (40% of customers) visits frequently but spends modestly (avg €33 per ticket). A €5 lift via deli pairings, applied to every Loyal Regular ticket, projects an annual revenue uplift in the order of low-single-digit percent of total revenue — directly executable through the 'Pair It' card at the till.
- 3. The store is perceived as a wine shop, not a delicatessen.** Olives dominate the deli mix; only 49% of customers buy premium deli (cheese, Spanish ham, salmon*), and no single segment makes premium its default choice. Eye-level pairing displays and trained staff suggestions are the lowest-cost lever available.

What this report covers: the customer profile (who walks in), the three segmentation lenses (revenue, behavior, spend tier), monthly-spend patterns including a counter-intuitive education finding, and the prioritised marketing action plan. All numbers are reproducible by re-running the dashboard on the live URL above.

**"Salmond" appears in the raw survey as a typo for Salmon and is displayed as "Salmon*" throughout the dashboard.*

Who Walks Through the Door

The typical customer is a **31–40-year-old male**, university-educated, who visits the store 1–2 times per week, drinks at parties or social gatherings, and pays with cash. Average basket: **€56**. The base skews mildly male overall but the picture is more nuanced when split by age band, gender and drinking occasion. Two demographic charts below frame the composition.

Population pyramid · 31–40 skews male (76M vs 55F); 41–50 skews female (54F vs 40M); +50 skews male again. The most-male age band is also the highest-frequency.

The Strongest Behavioural Signal — Where They Drink × Gender

Diverging horizontal bars · sort order is by male/female skew per occasion. This is the single most actionable demographic finding in the dataset.

86% of *With your couple* drinkers are male. **80%** of *Birthday party* drinkers are male. **73%** of *Restaurant* drinkers are female. These are not subtle differences — they are the strongest behavioural signals in the data.

In-store displays and bundle promotions should reflect this: anniversary / couple bundles signposted toward male shoppers, gift-ready bundles around birthday occasions, and a curated restaurant-style wine card visible to female shoppers.

Three Lenses, One Customer Base

We segment the same customers three ways, each answering a different question. No single lens is sufficient — the manager benefits from holding all three.

Lens	Method	Question it answers	Output
FM Segments	Median split on visit frequency × avg ticket	Who is worth what?	4 personas: Champions, Loyal Regulars, Occasion Splurgers, Casual Visitors
Behavioural K-Means	K-Means K=4 on (visits/mo × sociality)	How do they buy?	4 archetypes: Social Regulars, Daily Home Drinkers, Occasion Celebrants, Casual Home Drinkers
Spend Tiers (K-Prototypes)	Mixed-type clustering: Hamming on cats + Euclidean on Ticket	How much do they buy?	3 tiers: Entry / Core / Premium Buyers (anchored on ticket value)

Where Each FM Segment Sits

Customer share (left) vs revenue contribution (right) by FM segment. Champions: 37% of customers, ~68% of revenue. The single highest-priority group.

The FM and spend-tier lenses agree directionally: high-monetary FM segments (Champions + Occasion Splurgers) correspond closely to the Premium spend tier from K-Prototypes — independent confirmation that ticket value is a real differentiator. The behavioural K-Means lens then adds the missing piece: how to reach each group through a marketing channel that matches their lifestyle pattern.

On the Behavioural Segments tab, the deli mix is now expressed as a **Premium vs Entry** split per behavioural group — a cleaner read for campaign targeting than the eight-product breakdown.

Where the Money Actually Goes

We track **monthly spend** (ticket × visits per month), not just per-visit ticket. A daily €20 customer (€600/mo) is more valuable than a once-a-month €80 customer (€80/mo) — using monthly spend as the KPI reveals patterns that per-ticket views would hide.

Counter-intuitive: Education Is Not Predictive

Avg monthly spend per customer · Gender × Education tier. Darker = higher spend.

Two findings the marketing copy needs to absorb: (a) **Female · Basic-tier** customers are the highest-spending demographic in the entire base (€850/month, n=49); (b) **Male · University-educated** customers — the demographic most marketing copy implicitly assumes is 'premium' — are among the lowest (€446/month). Education is **not** a useful lever for targeting. Combining age, occasion and gender produces sharper signals.

The Marketing-Actionable Matrix

Avg monthly spend per customer · Age × Drinking occasion. The matrix to design themed promotions around — each cell is directly executable in a campaign. Visible on the Customer Behaviour tab.

Unlike age or education alone, drinking occasion translates directly into campaign mechanics: themed party packs for the *Parties* column, gift-ready bundles for *Birthday party*, restaurant wine cards for the *Restaurant* column, romantic boxes for *With your couple*. The highest-spending well-populated cohort (n ≥ 10) is 23–30-year-olds drinking at home — a candidate for a midweek 'tonight's pairing' shelf-talker programme.

Strategic Action Plan

Priorities are ranked by combined revenue impact and execution difficulty. The full per-segment playbook (4 actions per segment, two lenses) is available on the Action Plan tab of the live dashboard.

Priority	Segment / Driver	Action
P1	Champions 37% / 68% rev	Launch a named loyalty programme ('Cave Club'), private tastings, premium hamper bundles as default upsell. Retention-sensitive group — every lost Champion costs disproportionately.
P2	Loyal Regulars 40% / 30% rev	'Pair It' deli card at the till — one suggested deli match per wine. Monthly curated box (wine + 2–3 deli items) at a slight discount. Highlight premium deli provenance at eye level.
P3	Deli perception gap cross-cutting	Wine + deli pairing display at eye-level. Staff trained to suggest one product match per purchase. Restaurant-style wine card visible to female shoppers (73% of restaurant-occasion drinkers).
P4	Occasion Splurgers 13% / minor rev	Seasonal hamper range €60–100 (Christmas, Easter, summer). Social-media-ready gift packaging. WhatsApp campaign before Valentine's, Mother's Day, festive season.
P5	Casual Visitors 10% / minor rev	Instagram activation with entry-level pairing content. 'First deli taster' with any wine purchase. Monthly Saturday discovery tasting. Re-evaluate if no migration within 6 months.

What-If Projection

The dashboard's interactive simulator lets the manager combine three interventions and see the projected annual revenue impact in real time:

1. 20% of Casual Visitors matched to Loyal-Regulars-level visit cadence (activation marketing) +
2. €5 ticket lift for every Loyal Regular ('Pair It' deli upsell) +
3. 10% of Loyal Regulars elevated to Champions-level ticket (loyalty programme)

Projects a **+8.6% annual revenue uplift** (€2.78M → €3.02M) — directional only, based on the survey sample. Real impact would be confirmed against POS data.

Also Checked, Briefly

We verified — and chose not to highlight on the dashboard's front line — the relationships between payment mode and other attributes (weakest signal in the data; no clear marketing lever), the Cramér's V matrix across all categorical attributes (top pair scores 27/100, moderate), an unsupervised K-Prototypes spend-tier validation of the FM framework (available in the Behavioral Segments tab via the lens toggle), and several alternative chart formats (boxplots, parallel categories) that were rejected for lower insight density.

Note on the dataset: this is a synthetic dataset created for our course *Advanced Programming with Python*. Some rows contain unrealistic combinations — the segmentation methodology would be re-fit on real POS data when available.

For interactive exploration, filter controls and the live what-if simulator:

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